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Review Article

The effects of computerized executive function training on the youth with ADHD: A systematic review and meta-analysis

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ABSTRACT

Attention-deficit/hyperactivity disorder (ADHD) is one of the most prevalent neurodevelopmental disorders among the youth, imposing significant stress and economic burdens on so many countries and families. With the widespread adoption of computers, computerized executive function training (CEFT) has emerged as a potentially safer, more convenient, and accessible treatment of ADHD. However, the efficacy of CEFT on the youth with ADHD is debated. To evaluate the efficacy of CEFT for youth with ADHD, we examined its immediate, transfer (including near transfer to other cognitive functions and far transfer to ADHD symptoms, academic performance, and quality of life), and sustained effects. We systematically searched four databases: PubMed, Web of Science, CNKI (China National Knowledge Infrastructure), and Wanfang Data, with search terms encompassing ADHD, CEFT, and youth. After screening, thirty-six articles with 1798 participants have been identified. In the included studies, participant sample sizes varied from 17 to 104 individuals, with mean age spanning from 5 to 15.6 years. Study quality and risk of bias were assessed using the EPHPP tool, with 22 % of studies ($n = 8$) at low risk, 56 % ($n = 21$) at moderate risk, and 22 % (n = 8) at high risk. The meta-analysis results revealed significant immediate effects, near transfer effects, far transfer effects, and sustained effects. The meta-regression was conducted but no significant moderator was found. Our study indicates considerable efficacy of CEFT to youth with ADHD, and further exploration of potential moderators may be beneficial to refine its application.

1. Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is one of the most prevalent neurodevelopmental disorders among the youth (Perou et al., 2013), with a global prevalence estimated between 2 % and 7 % (Chan et al., 2024; Elia et al., 2018). ADHD is characterized by symptoms of inattention, hyperactivity, and impulsivity, which can have detrimental effects on children's academic performance and social interactions. Beyond the physical and psychological distress experienced by those diagnosed, ADHD also entails a substantial economic burden on families. Research conducted in Germany revealed that patients with ADHD incur an average of 1500 euros more in annual medical expenses than their non-ADHD peers (Libutzki et al., 2019). They also documented that, for children with co-occurring conditions, such as mood disorders, substance abuse, and obesity, the additional costs can amount to 2800 euros per year.

The primary treatments currently employed for the youth with ADHD consist of pharmacotherapy, widely regarded as the cornerstone of treatment for ADHD due to its robust efficacy and safety (Bellato et al., 2025; Ching et al., 2019; Storebø et al., 2015), and cognitive-behavioral therapies (Sibley et al., 2023). Although these interventions effectively ameliorate ADHD's core symptoms, issues such as low medication adherence and parental concerns regarding potential side effects pose significant challenges (Meyer et al., 2020). Moreover, cognitive-behavioral therapies can be prohibitively expensive due to a lack of qualified professionals, further complicating access to effective treatment for ADHD. Meanwhile, cognitive rehabilitation methods have emerged as an accessible and potentially safer alternative for treating

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ADHD and other psychiatric disorders. For instance, Buonocore et al. (2015) evaluated a combined intervention of computerized cognitive remediation and metacognitive training in adults with schizophrenia, demonstrating significant improvements in cognitive flexibility and social functioning. Hoorelbeke et al. (2015) investigated a computerized cognitive training (CCT) program targeting cognitive control in patients with depression, finding reductions in stress reactivity and rumination alongside enhanced cognitive control. Recent systematic reviews further support the efficacy of cognitive remediation in psychiatric disorders. For instance, Vita et al. (2021) analyzed 130 studies from 2011 to 2020 involving 8851 participants, confirming that cognitive remediation significantly improves cognition and functioning in schizophrenia. Similarly, Mokhtari et al. (2023) emphasized the value of integrating cognitive rehabilitation into clinical practice, demonstrating notable reductions in core symptoms of major depressive disorder (MDD), highlighting its potential as an adjunctive approach for youth with MDD and other mental disorders. With a notable focus on executive functions training, CCT may serve as a potentially effective treatment for patients with ADHD (Westwood et al., 2023).

Executive functions (EFs) encompass the cognitive processes through which individuals regulate and manage their thoughts and behaviors, incorporating higher-order cognitive functions, including working memory, inhibitory control, cognitive flexibility, and sustained attention (Ferreri et al., 2022). Extensive literature supports a robust association between ADHD symptoms and deficits in executive functioning (Kofler et al., 2020; Wu et al., 2022). Computerized executive functions training (CEFT) involves utilizing software and hardware technologies to design systematic, individualized, and quantifiable tasks and intervention programs that address the EFs weaknesses in children with ADHD (Luo et al., 2022; Rapport et al., 2013). Through repeated training, these interventions aim to enhance self-control, working memory, cognitive flexibility, and thereby improve issues related to attention deficits, hyperactive behavior, social adaptation, and learning difficulties.

CEFT for youth with ADHD targets four key domains: working memory, cognitive flexibility, inhibitory control, and sustained attention (Barkley, 1997; Gibson et al., 2011; Sergeant et al., 2002; Willcutt et al., 2005). For instance, Mawjee et al. (2017) randomized 38 children with ADHD into three groups: standard-duration (45 min/session, 25 sessions), short-duration (15 min/session, 25 sessions), and control. Both intervention groups showed significant improvements in working memory and hyperactivity compared to controls. Similarly, Dentz et al. (2020) used Cogmed software for working memory training, reporting enhanced attention and overall functioning. Meyer et al. (2020) adapted a Stop-Signal Task to improve inhibitory control, with Electroencephalogram (EEG) data showing reduced ADHD symptoms, linked to changes in theta band energy and N200 component latency/amplitude. Tamm et al. (2014) conducted an eight-week training with 24 children, improving cognitive flexibility and reducing attentional issues. Korpa et al. (2020) implemented a 20-session protocol targeting working memory, inhibitory control, and sustained attention, yielding improvements in EFs and ADHD symptoms. Rosa et al. (2020) used Functional magnetic resonance imaging (fMRI) to compare brain activation in children with ADHD and typically developing peers, finding that training enhanced sustained attention, associated with changes in parietal cortex and striatal-limbic activation.

Although numerous studies suggest that CEFT can effectively enhance EF deficits and behavioral outcomes in children with ADHD, some research reports inconsistent results. Dongen-Boomsma et al. (2014) trained 47 children with ADHD using Cogmed working memory tasks over five weeks, employing the ADHD Rating Scale IV (ADHD-RS) alongside assessments of neurocognitive function, daily EFs, and clinical functioning, but found no significant effects on behavior, neurocognition, or daily executive abilities. Similarly, Woltering et al. (2021) randomized 89 children with ADHD into long-duration (45 min/session, 25 sessions), short-duration (15 min/session, 25 sessions), or control groups, finding no notable improvements in neurocognitive or behavioral transfer effects compared to controls. Bikic et al. (2021) evaluated 70 children with ADHD, noting that an eight-week CEFT combined with standard treatment did not significantly improve sustained attention. Likewise, Dovis et al. (2015) found no ADHD symptom improvements in 89 children after game-based CEFT.

With respect to the effectiveness of CEFT on the youth with ADHD, researchers not only debate the immediate effects of these interventions but also lack consensus regarding their transferability and sustainability. Transferability refers to the ability of intervention effects to generalize to other contexts (Saccardo et al., 2024), while sustainability denotes the durability of effects over time (Stanford et al., 2023), both of which are critical indicators of intervention efficacy. In terms of transferability, Bigorra et al. (2016) found evidence of far transfer effects from CEFT, whereas Ackermann et al. (2018) reported that cognitive intervention effects did not extend beyond the targeted domains. Furthermore, Jones et al. (2020), in their n-back computerized intervention study, observed that while training effects were transferable within similar tasks, they failed to translate into broader contexts. Regarding sustainability, Donk et al. (2015) found no enduring effects from computerized executive function interventions, while Kofler et al. (2020) reported that improvements from inhibitory control training were maintained at a 2–4 months follow-up. Dentz et al. (2020) similarly noted that the effects of computerized working memory interventions persisted for up to 6 months after intervention.

In light of the variability in findings across studies, recent systematic reviews and meta-analyses of this growing literature have provided preliminary evidence that CEFT can improve various outcomes of children and adolescents with ADHD, including working memory, reading comprehension, and some clinical symptoms (Dhir et al., 2021; Westwood et al., 2023). However, these meta-analyses primarily focused on the impacts of training on specific EF components without delineating differences among various forms of executive function training. Furthermore, there has been insufficient examination of the sustainability of intervention effects, and the impact of critical moderating variables, such as whether the intervention is gamified or the intensity of the intervention, remains to be clarified.

Therefore, the present study aims to address the ongoing debate regarding the efficacy of CEFT on youth with ADHD, and explore potential factors influencing the effectiveness of training. Specifically, we will systematically review existing trials of CEFT on the youth ($<$ 18 years old) with ADHD, and apply meta-analytic techniques to establish the effects of CEFT on the youth with ADHD and the impact on training effects of potential moderators, such as the training target, training intensity, gamification, and so on. We also sought to investigate 1) the transfer effect (general cognitive function, clinical symptoms, academic performance, and quality of life), and 2) the sustained effect of CEFT on the youth with ADHD.

Based on previous research, we propose the following hypotheses. First, CEFT can effectively enhance cognitive abilities of the youth with ADHD. Second, CEFT can significantly alleviate ADHD symptoms, thereby improving their academic performance and quality of life (transfer effect). Third, the intervention effects of CEFT on the youth with ADHD would exhibit a degree of sustainability (sustained effect). Fourth, the effectiveness of CEFT may be influenced by various moderating factors, particularly the EF components of the training program.

2. Methods

2.1. Search Strategy

The current study is in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (Liberati et al., 2009). Articles in both English and Chinese from PubMed, Web of Science, CNKI (China National Knowledge

Infrastructure), and Wanfang Data (a comprehensive Chinese-language database) were searched until May 2025. The following search terms were utilized in titles, abstracts, table of contents, indexing, and key concepts: (ADHD OR attention deficit hyperactivity disorder OR hyperactivity disorder) AND (children OR child OR adolescent OR adolescents OR development OR developmental OR school-age OR underage OR lifespan) AND (computer OR computerized OR digital OR digitized) AND (cognitive control OR executive control OR executive function OR working memory OR inhibition control OR cognitive flexibility OR sustained attention) AND (training OR intervention).

2.2. Exclusion criteria

Among all the retrieved studies, those meeting the following criteria were excluded from this meta-analysis: 1) duplicates; 2) review or meta-analysis studies; 3) case studies; 4) studies not involving computer-based training or interventions; 5) studies not including an ADHD group; 6) studies only including adults (age $\geq$ 18); 7) studies with effect sizes that could not be extracted or calculated; 8) studies not in English or Chinese; 9) non-peer-reviewed articles; 10) studies including comorbid conditions (other mental disorders); 11) studies not targeting executive function training.

Two authors independently reviewed the titles and abstracts of identified studies, as well as any eligible full-text articles. Discrepancies were resolved through discussion with a third researcher to achieve consensus on the final coding.

2.3. Data extraction

The overall quality of all studies was evaluated using the Effective Public Health Practice Project (EPHPP) Quality Assessment Tool for Quantitative Studies (Effective Public Health Practice Project, 1998). The EPHPP Quality Assessment Tool evaluates study quality based on eight components: 1) selection bias; 2) study design; 3) confounders; 4) blinding; 5) data collection methods; 6) withdrawals and dropouts; 7) intervention integrity; and 8) analysis. A positive rating for each component indicates a low risk of bias, while a negative rating suggests a high risk. The EPHPP tool was chosen for its established framework for systematically assessing the quality of included studies.

The data extracted from the articles and the coding of study characteristics included: the first author, publication year, language of article, mean age (years) of participants, sample size, gamification of training program, training target, training intensity, type of control group, near transfer effects, ADHD symptoms, assessment format of ADHD symptoms, academic achievement, quality of life, and sustained effects (follow-up).

Following the recommendations of Higgins (2009), two authors of the present study preliminarily coded the literature and verified their results independently. Any discrepancies were resolved through consultation with a third researcher to reach a consensus on the final coding.

2.4. Outcomes

For outcomes, we adopted a three-tiered approach to examine the intervention effects, which included the immediate effects, transfer effects, and sustained effects of CEFT in youth with ADHD. Specifically, we first distinguished the performance of EFs, which effectively reflected the immediate effects of the intervention. Regarding secondary outcomes, we examined the transferability of the intervention's impacts. We hypothesized that other cognitive functions of participants, as well as academic performance, ADHD symptoms, and quality of life, would improve if the intervention effects were transferable. Furthermore, we regarded the sustainability of the intervention effects as an important outcome of interest.

When multiple intervention outcomes were available in a study, we prioritized EF outcomes as the primary outcome for effect size calculation. If the primary outcome did not adequately reflect the intervention effect, we combined the multiple intervention outcomes using a hierarchical method (Borenstein et al., 2021) to create a synthetic variable and subsequently included the combined effect size in the overall analysis.

2.5. Statistical analysis

The analyses were conducted using Comprehensive Meta-Analysis (CMA) Version 3.3 (Borenstein, 2022). We performed a random-effects meta-analysis using the standardized mean difference, specifically Hedges' g (Hedges and Olkin, 1985), as the effect size for the experimental group compared to the control group. Effect sizes were interpreted as follows: 0.2 indicates a small effect size, 0.5 indicates a medium effect size, and 0.8 indicates a large effect size. The effect size Hedges' g was calculated using the sample sizes of the control and experimental groups, as well as the post-test means and standard deviations. In cases where means or standard deviations were unavailable, the data were transformed or selected for computation using χ^2, t, or F statistics.

To assess whether heterogeneity was due to sampling error, between-study heterogeneity was measured using Cochran's Q and Higgins' I^2 statistics. Here, I^2 indicates the proportion of variance among studies relative to the total variance (I^2 = 25 %, 50 %, 75 % signifies low, moderate, and high heterogeneity, respectively). A significant Q and I^2 $\geq$ 75 % indicated substantial heterogeneity between studies, suggesting that the use of a random-effects model was appropriate (Huedo-Medina et al., 2006).

2.6. Publication bias

Several methods were utilized to evaluate potential publication bias. Funnel plots and Egger's test were used first to test for publication bias. An intercept close to 0 and non-significant in Egger's linear regression indicated a lower likelihood of publication bias (Egger et al., 1997).

Furthermore, the fail-safe N method was used if both Egger's test and funnel plots indicated significant publication bias (Khoury et al., 2013; Rosenthal and Bulletin, 1979). The fail-safe N, the minimum number of studies with non-significant results needed to overturn the meta-analysis conclusions, was then calculated. A larger fail-safe N indicated a lower likelihood of bias; however, a fail-safe N less than 5 k + 10 (where k is the number of original studies) suggested that publication bias was a concern (Rothstein et al., 2006).

2.7. Robustness and sensitivity analysis

Considering the results of the meta-analysis were potentially influenced by factors such as inclusion criteria, data extraction methods, and handling of missing values, we performed a sensitivity analysis on the overall effect size estimate by re-evaluating effect sizes after excluding outliers.

2.8. Moderator analysis

We followed the PICO principle (Akobeng, 2005), focusing on three aspects: 1) participant characteristics, 2) intervention, and 3) outcomes. Based on evidence-based medicine principles, we identified moderators which might affect the effectiveness of computer-based interventions for EFs in youth with ADHD. These included: first, participant characteristics, such as age and sample size; second, intervention factors, including gamification of the training program, control group type, training target, and training duration; and third, outcomes, specifically the number of EFs targeted by the intervention.

The meta-regression analysis using linear mixed-effects models was conducted to further examine the impact of potential moderators on

intervention effectiveness. Before including categorical variables in the regression analysis, one level of each variable was chosen as the reference group, while other levels were automatically generated as dummy variables by CMA Version 3.3 (e.g., if the regional variable was divided into east or west categories, west could be designated as the reference group, with east transformed into a dummy variable); the choice of reference variable was not expected to affect the model's results.

3. Results

3.1. Study selection

Systematic searches of peer-reviewed databases up to May 2025 yielded 1639 articles. After removing duplicates, 1311 articles were screened. The characteristics of final included studies are as follows: Overall, 36 articles (with 37 studies) including 1798 participants were suitable for meta-analysis (Fig. 1). In the 37 studies, sample sizes ranged from 17 to 104, mean ages ranged from 5 to 15.6 years, and the proportion of female participants ranged from 0 to 1. Studies in English (92 %, $n = 34$) outnumbered those in Chinese (8 %, n = 3), and Western studies (65 %, $n = 24$) outnumbered Eastern studies (35 %, $n = 13$). Active control groups were used in 16 studies (43 %), non-active control groups in 15 studies (41 %), and other alternative control groups in 6 studies (16 %). Most interventions (73 %, $n = 27$) were gamified, while 27 % ($n = 10$) were non-gamified. Most interventions (81 %, $n = 30$) lasted less than 10 weeks, 16 % ($n = 6$) lasted 10 to 20 weeks, and 3 % (n = 1) exceeded 20 weeks, with an average duration of 7.27 weeks. Among studies, 8 % (n = 3) addressed all four EF aspects (working memory, inhibitory control, cognitive flexibility, and sustained attention), 24 % ($n = 9$) addressed three, 19 % ($n = 7$) addressed two, and 49 % ($n = 18$) focused on one. Study details are in Supplementary Table S1. Over half of the studies were rated "Moderate" ($n = 21$), with "Strong" and "Weak" ratings each at 22 % ($n = 8$). Quality assessment details are in Supplementary Table S2.

3.2. Immediate effects

A total of 35 effect sizes comprising 1618 participants were included in the meta-analysis of immediate effects. The pooled effect size was moderate and statistically significant ($g = 0.488$, 95 % CI [0.371, 0.606], $p < 0.01$) (Fig. 2), with non-significant between-study heterogeneity ($Q = 45.759$, $\tau^2 = 0.031$, $p > 0.05$, $I^2 = 25.697$ %). The funnel plot showed no significant asymmetry (Supplementary Fig. S1), indicating negligible publication bias. Egger's test was also non-significant ($Z = 0.630$, $p > 0.05$). Meta-regression analyses exploring potential moderators yielded no significant findings. A leave-one-out sensitivity analysis confirmed the robustness of the results, identifying no outliers.

3.3. Transfer effects

Transfer effects were assessed through two separate meta-analyses, focusing on academic performance and EFs not directly targeted in the studies. For near transfer effects, eight effect sizes involving 376 participants were analyzed, revealing a moderate pooled effect size ($g = 0.567$, 95 % CI [0.219, 0.914], $p < 0.01$) (Fig. 3). Significant between-study heterogeneity was observed ($Q = 18.688$, $\tau^2 = 0.151$, $p < 0.01$,

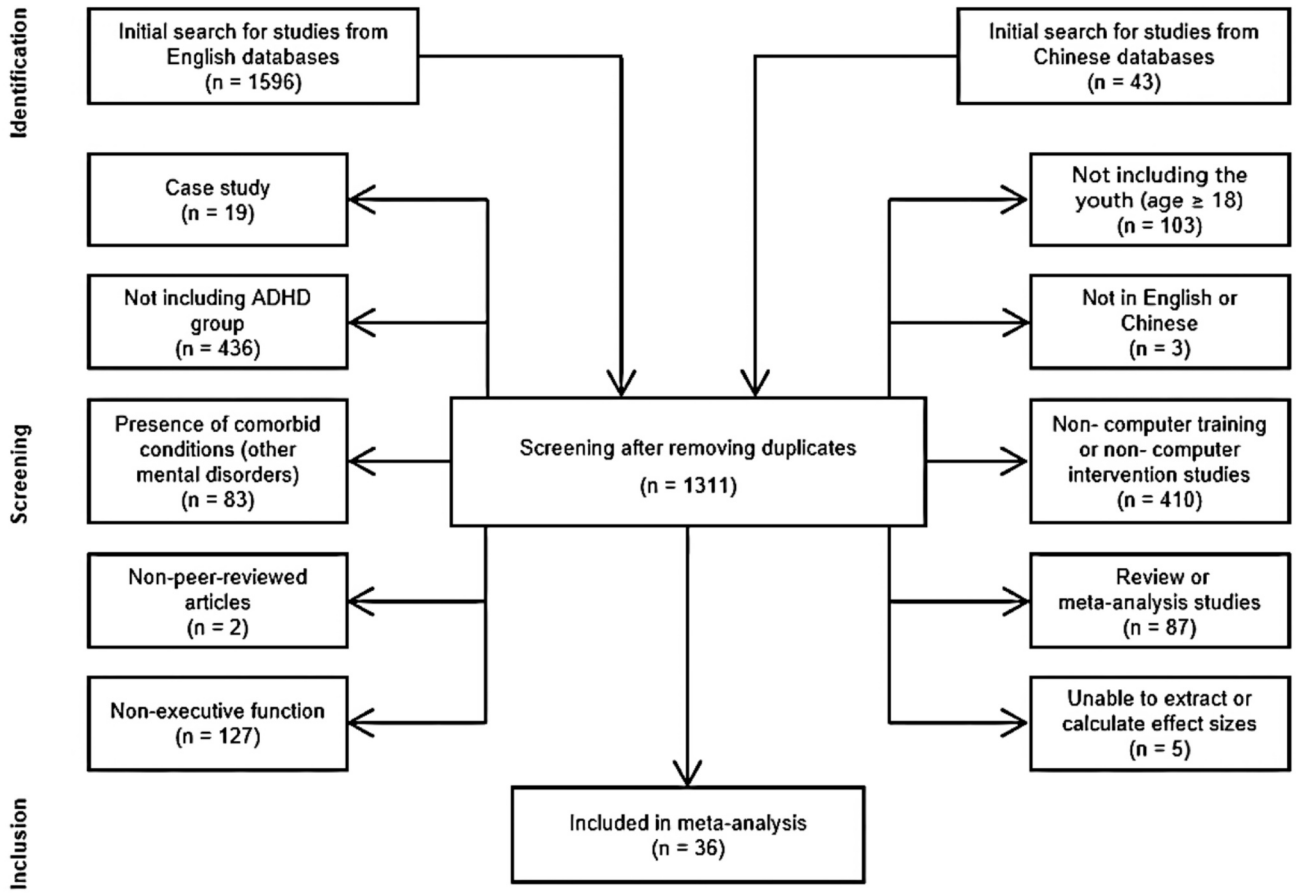


Fig. 1. The preferred reporting items for systematic reviews and meta-analyses (PRISMA) flowchart of the selection process for included articles.
Note. ADHD = Attention-Deficit/Hyperactivity Disorder.

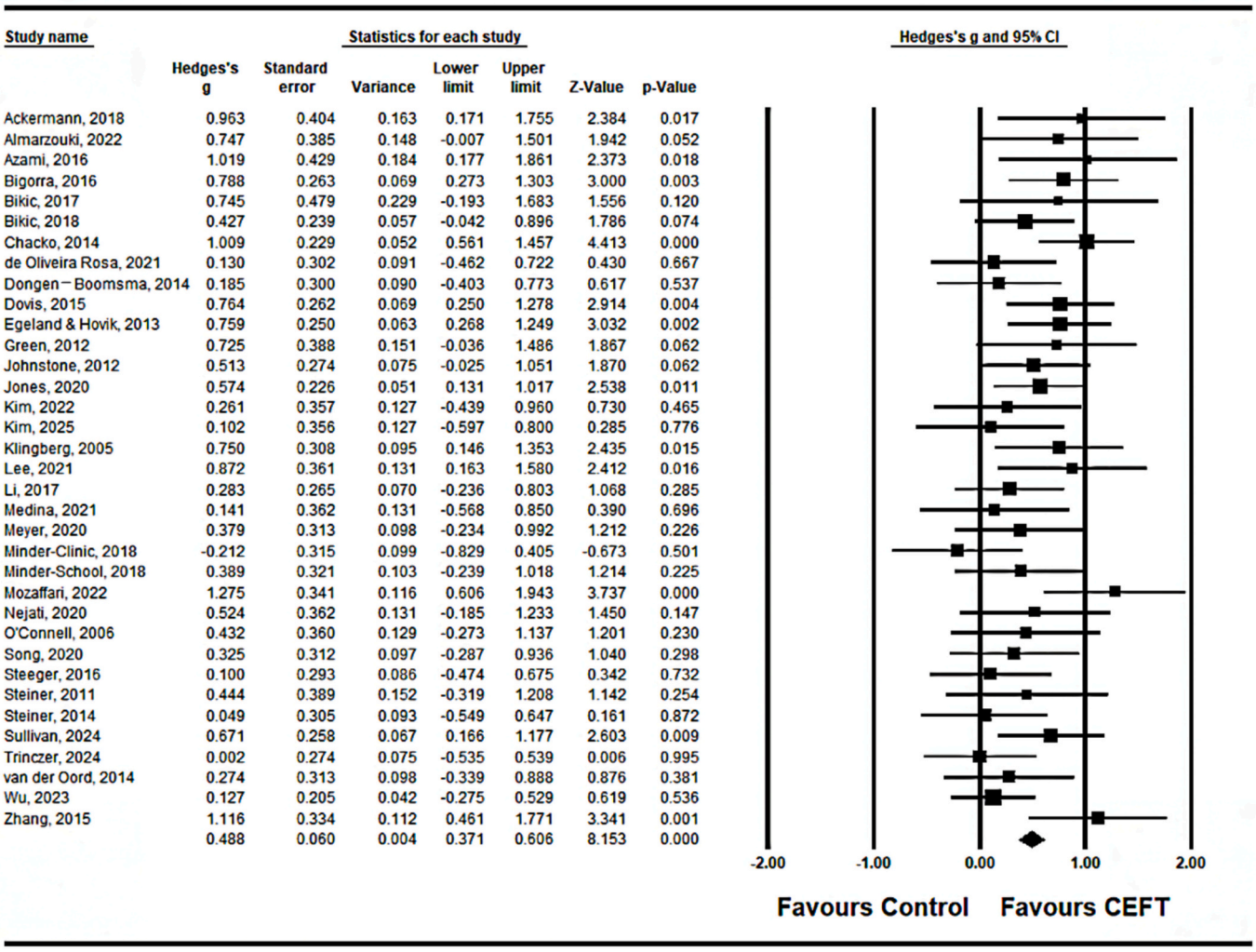


Fig. 2. The forest plot of immediate effects.
Note. CEFT = Computerized Executive Functions Training.

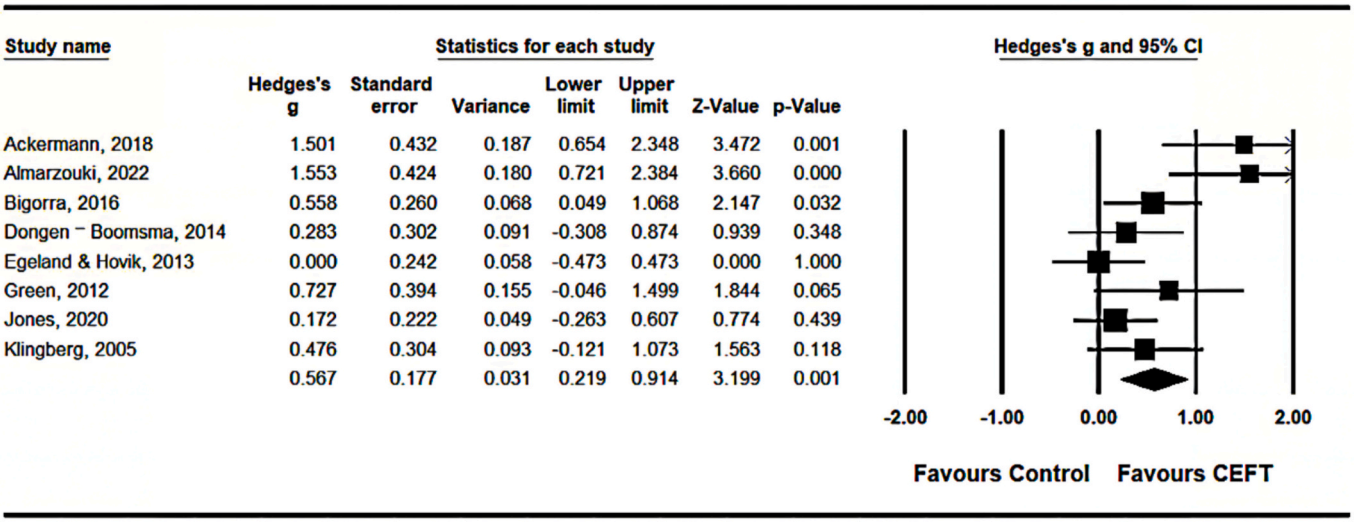


Fig. 3. The forest plot of near transfer effects.
Note. CEFT = Computerized Executive Functions Training.

$I^2 = 62.543$ %), but no moderators significantly explained this heterogeneity. Funnel plot asymmetry (Supplementary Fig. S2) and Egger's test ($Z = 4.61$, $p < 0.01$) suggested potential publication bias. The Classic Fail-safe N was 46, slightly below the recommended threshold of 50. Nonetheless, sensitivity analysis confirmed the robustness of these findings, indicating that CEFT may enhance participants' quality of life and general cognitive abilities.

For far transfer effects, eight effect sizes involving 464 participants were included, showing a moderate pooled effect size ($g = 0.468$, 95 % CI [0.135, 0.802], $p < 0.01$) (Fig. 4) with moderate heterogeneity ($Q = 20.216$, $\tau^2 = 0.145$, $p < 0.01$, $I^2 = 65.374$ %). Funnel plot asymmetry (Supplementary Fig. S3) and Egger's test ($Z = 8.13$, $p < 0.01$) suggested possible publication bias, indicating potential overestimation of the effect size. The Classic Fail-safe N was 35, below the recommended threshold of 50, further highlighting publication bias concerns. However, sensitivity analysis demonstrated considerable robustness, suggesting that while caution is warranted, the results remain stable. Small sample sizes and the limited number of studies may contribute to the observed publication bias.

3.4. Sustained effects

Sustained effects were evaluated through EFs measures, with eight effect sizes involving 428 participants. The pooled effect size was moderate ($g = 0.541$, 95 % CI [0.308, 0.774], $p < 0.01$) (Fig. 5). Between-study heterogeneity was not significant ($p > 0.05$), so moderator analysis was not conducted. Slight funnel plot asymmetry was observed (Supplementary Fig. S4), but Egger's test was non-significant ($p > 0.05$). Sensitivity analysis confirmed the robustness of these findings.

3.5. Effects of moderators

Meta-regression analyses employing mixed-effects model were performed to investigate the influence of potential moderators on training outcomes. Specifically, we evaluated the effects of age, gender, control group type, training program, and assessment methods. None of these moderators exhibited statistically significant effects, suggesting a limited impact on the overall training outcomes.

4. Discussion

To examine the effectiveness of CEFT for youth with ADHD, we conducted a meta-analysis and systematic review focusing on three primary outcomes: immediate effects, transfer effects, and sustained effects. Our findings demonstrate that CEFT significantly improves ADHD symptoms, enhances quality of life, and bolsters general EFs among youth with ADHD. Furthermore, the training effects appear to persist over extended periods, suggesting that CEFT has considerable potential for broader applications, including clinical therapy, educational training, and preventive interventions.

The current study confirms that CEFT improves EFs in youth with ADHD, aligning with prior findings (Sonuga-Barke et al., 2013). However, CEFT's impact varies across EFs domains. Cortese et al. (2015) found no significant improvements in inhibitory control or cognitive flexibility, though working memory and sustained attention benefited. Similarly, Gates et al. (2019) reported modest short-term EFs gains, noting that low methodological quality in studies warrants cautious interpretation. Notably, CEFT appears more effective for children from lower socioeconomic backgrounds than higher ones (Scionti et al., 2019), supporting Hackman et al. (2015) framework and highlighting the potential of CEFT to promote equitable educational outcomes.

Significant transfer effects of CEFT were also identified in the present meta-analysis. Previous studies have consistently reported robust near transfer effects (Melby-Lervag et al., 2016; Scionti et al., 2019), and our results corroborate these findings. In contrast, far transfer effects are less frequently observed. Cortese et al. (2015) noted transfer to EFs and mathematical ability, though with a small effect size for the latter. Similarly, Luis-Ruiz et al. (2020) reported near transfer effects but noted that these benefits did not extend to more complex, real-life scenarios. Although our meta-analysis revealed a significant and moderate far transfer effect ($g = 0.515$, 95 % CI [0.129, 0.901], $p < 0.01$), this finding should be interpreted with caution due to limitations, including small sample sizes and a limited number of included studies. Consequently, the far transfer effects of CEFT require further investigation to establish their robustness.

Stronger near transfer compared to far transfer in CEFT may arise from the interplay of general and specific EF mechanisms. Fu et al. (2022) found similar single-neuron activation patterns and overlapping geometric representations in the dorsal anterior cingulate cortex (dACC) across Stroop, task-switching, and working memory tasks. Similarly, Morales et al. (2018) showed cross-task activation in the anterior

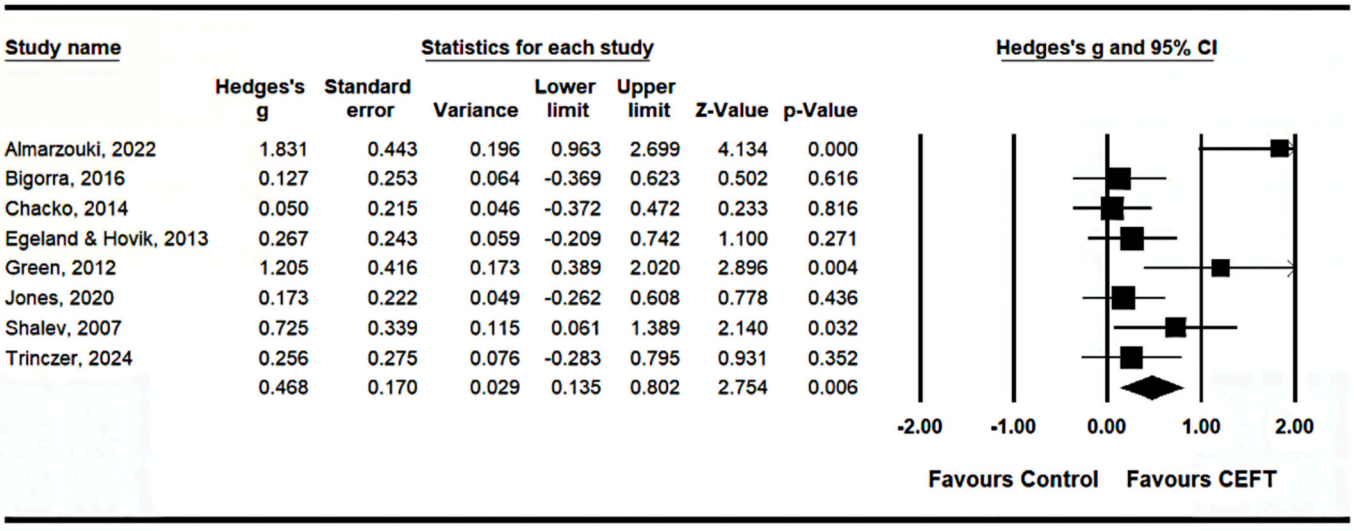


Fig. 4. The forest plot of far transfer effects.
Note. CEFT = Computerized Executive Functions Training.

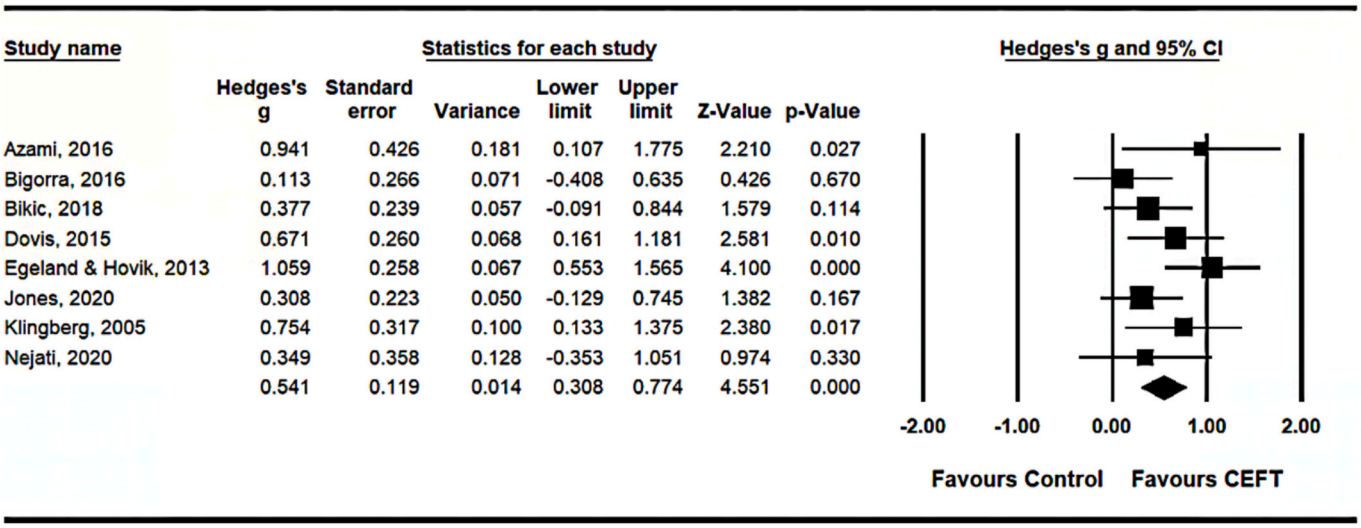


Fig. 5. The forest plot of sustained effects.
Note. CEFT = Computerized Executive Functions Training.

cingulate cortex (ACC) and medial prefrontal cortex (mPFC), supporting their role in domain-general cognitive control. These findings suggest shared neural mechanisms facilitate CEFT's transfer to related cognitive domains. Conversely, Soutschek and Schubert (2013) demonstrated domain-specific control, with working memory load impairing conflict regulation in a non-emotional Stroop task, while an emotional Go/No-go task only disrupted the emotional Stroop task. Ko et al. (2021) used fMRI to show the prefrontal cortex (PFC) encodes task-specific scene attributes (e.g., temperature, sound level) only when relevant, while scene category representations remain invariant. Badre (2025) proposed a model where cognitive control dynamically adapts to task demands, explaining why CEFT effects transfer more readily to similar EFs (near transfer) but less to complex scenarios (far transfer).

Regarding the duration of CEFT effects, the majority of studies included in this review reported that benefits persist over the long term. These findings align with previous research. For instance, Westwood et al. (2023) observed sustained improvements in EFs, such as verbal working memory, at six months post-intervention. Liu et al. (2024) similarly reported enduring effects on ADHD symptoms and inattention, although improvements in hyperactivity symptoms were not sustained. An umbrella review of 16 studies explored factors contributing to long-term effects of computerized training, concluding that both intervention frequency and duration influence sustained outcomes, with longer interventions linked to more robust effects (Türk et al., 2023). Another meta-analysis highlighted intervention duration as a critical factor for sustained improvements in EFs, including organizational skills and attention, among children with ADHD (Bikic et al., 2017). These findings suggest that extending the duration of CEFT could yield more stable and pronounced effects.

The results of current meta-analysis demonstrated the significant and sustained efficacy of CEFT for patients with ADHD, highlighting its potential as a therapeutic intervention. For instance, Rosa et al. (2017) investigated CEFT in ADHD patients who continued stimulant medications (e.g., methylphenidate and lisdexamfetamine) during training. The study reported a significant reduction in parent-rated ADHD symptoms post-intervention, with no reported side effects, suggesting CEFT's compatibility with pharmacotherapy. Similarly, Luo et al. (2022) combined neurofeedback (NFT) with CEFT in a three-month intervention for children with ADHD, observing significant improvements in hyperactive/impulsive symptoms, EFs, and daily life skills. Beyond ADHD, CEFT shows potential in treating other psychiatric disorders characterized by executive dysfunction and PFC abnormalities. Calkins and Otto (2013) found that improved performance during CEFT correlated with reduced symptom severity, indicating CEFT's potential to mitigate OCD symptoms. The authors propose that CEFT's efficacy in OCD may derive from enhanced PFC activation, a region critically implicated in the disorder's core symptoms, including intrusive thoughts and compulsive behaviors (Goodman et al., 2021). Similarly, systematic reviews suggest that CEFT holds promise for other psychiatric and mental disorders characterized by EF deficits and PFC dysfunction. For instance, Pauley et al. (2023) reported that digital cognitive training bolster cognitive control in anxiety disorders, enhancing attentional focus and inhibitory processes. In addition, Goldberg et al. (2023) showed that cognitive training fosters cognitive flexibility and stabilizes mood in mood disorders like bipolar illness. Furthermore, Hanegraaf et al. (2021) demonstrated that cognitive training alleviates executive deficits in borderline personality disorder. Given its broad potential, clinicians are recommended to consider CEFT as an adjunctive therapy, especially for youth.

The present meta-analysis did not identify significant moderators of CEFT efficacy, which is in consistent with some prior studies (Shou et al., 2022; Westwood et al., 2023). However, certain evidence suggests that training effects may be influenced by various factors. For example, Zou et al. (2024) conducted subgroup analyses and found that children under 10 years old exhibited greater training efficacy, indicating that age may modulate the effects of CEFT. Similarly, Chen et al. (2021) performed univariate meta-regression analyses and determined that training frequency significantly impacted EFs and ADHD symptoms, with higher training frequency potentially leading to greater CEFT efficacy. Additionally, a meta-analysis indicated that Asian patients showed greater improvements in inattention compared to European patients (He et al., 2023). This difference might be attributable to cultural variations in educational practices, where Asian contexts often emphasize disciplined, attention-focused activities, potentially enhancing responsiveness to CEFT.

Although our study did not identify significant moderators, we predicted that the training format and focus would significantly influence the efficacy of CEFT. Specifically, we hypothesized that the gamification of CEFT (e.g., game-based versus non-game-based training formats) might enhance efficacy by increasing participant engagement and motivation. We reasoned that children would exhibit greater engagement with gamified training formats. However, our results showed no significant effect of gamification, similar to findings from Vermeir et al. (2020). We speculate this may be due to both gamified and non-gamified CEFT formats being equally structured to target core cognitive

processes, resulting in comparable outcomes across training modalities. Similarly, we anticipated that the number of EFs targeted by CEFT (e.g., single versus multiple functions) might influence outcomes, with training focusing on fewer functions potentially yielding stronger effects. However, no significant moderation was observed, aligning with results reported by (Dovis et al., 2015). This could be attributed to the limited duration of individual training sessions, where targeting multiple EFs simultaneously reduces the time devoted to training each specific function, resulting in minimal overall improvement in training outcomes.

Additionally, we expect different control groups, especially those with active control measures, will impact the final evaluation of training efficacy. In our study, we conducted a meta-regression to examine the effects of active and non-active controls on training outcomes. Moreover, we specified active controls as medication and placebo training, but neither significantly moderated executive function outcomes or symptom severity, consistent with findings by Cortese et al. (2015) and Vita et al. (2021). In their meta-analysis of cognitive training for ADHD, Cortese et al. (2015) found no significant moderation by control group type and attributed this to active controls reducing rater bias, thereby producing results similar to objective assessments and reducing differences with non-active controls. Similarly, Vita et al. (2021) reported no significant moderation of control group categories in cognitive remediation (CR) for schizophrenia. The authors suggested that CR specifically targets cognitive performance, whereas other control conditions do not, and this specificity reduces the differential impact of control group type on outcomes. Although control group type does not significantly affect training outcomes, we consider active controls preferable to non-active controls in studies, as their similar training processes help blind raters and participants, thereby reducing rater bias and enhancing the reliability of efficacy estimates.

Despite these findings, the exploration of moderators remains critical to optimize CEFT's therapeutic potential and inform strategies for achieving more robust outcomes. Future research should prioritize high-quality and well-powered studies with greater variability in these factors to elucidate their roles in CEFT efficacy.

4.1. Limitations

Our meta-analysis clarifies the efficacy of CEFT for youth with ADHD, but two limitations merit attention. First, while most effect sizes showed low heterogeneity, transfer effects displayed moderate residual heterogeneity, not fully explained by moderators in our meta-regression. This suggests that factors such as individual cognitive profiles or training protocol variations may influence outcomes. Future studies should explore adaptive CEFT approaches tailored to specific ADHD presentations or co-occurring conditions to enhance transfer effects and provide clinicians with practical tools to improve intervention outcomes.

Second, some effect sizes were drawn from studies with small sample sizes, which may introduce variability and limit the precision of our pooled estimates. To build on these findings, future research should focus on large-scale, multi-site randomized controlled trials to validate and extend our results. These studies could include diverse populations and employ advanced neuroimaging or behavioral metrics to better understand the neural and cognitive changes driven by CEFT, paving the way for innovative treatments that combine CEFT with other therapeutic approaches to optimize outcomes for youth with ADHD.

5. Conclusion

This meta-analysis and systematic review provide robust evidence that CEFT significantly benefits minors with ADHD, particularly by enhancing EFs essential for daily life. Our findings confirm the immediate, transfer, and sustained effects of these interventions, resolving debates about their efficacy and offering a comprehensive synthesis of their impact. Theoretically, this study deepens understanding of how targeted cognitive training mitigates EF deficits in ADHD, refining models of neurocognitive development. Practically, it highlights computerized training as a scalable, engaging adjunctive tool in clinical settings, complementing traditional therapies. By improving cognitive outcomes, these interventions support academic performance and social integration for youth with ADHD, paving the way for more effective, evidence-based strategies to empower them.

CRediT authorship contribution statement

Shibo Yan: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Xiaoxiang He:** Writing – original draft, Methodology, Data curation. **Yifan Zhang:** Data curation. **Jing Yang:** Data curation. **Xiang Xu:** Data curation. **Xinyu Liang:** Data curation. **Ting Xu:** Data curation. **Wenjie Dou:** Writing – review & editing. **Baoming Li:** Writing – review & editing, Funding acquisition. **Zhenghan Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Transparency and openness

Ethics committee approval (2021–1357) was obtained from Ethics Committee for Scientific Research of School of basic medicine of Hangzhou Normal University. All materials related to this systematic review and meta-analysis are available upon reasonable request.

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Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.120730.

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